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Gradient-free concatenated fusion neural network for early diagnosis of Parkinson's disease using MRI data

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Abstract
Parkinson's disease (PD) is a neurological condition marked by significant motor deficits resulting from damage to brain cells. It is essential to utilise MRI data to predict PD at an earlier stage, as current research often focuses on late disease stages, which limits the efficacy of early diagnosis approaches. This research represents a new effort to apply a novel method for early PD diagnosis, utilising a vast database of MRI scans. This methodology proposes a new deep architecture, the Concatenated Fusion-based Neural Network Strategy (CFNN). This integrates CNN models with advanced image analysis methods that incorporate both structural and functional features of the brain. It also involves preprocessing and normalizing MRI data to extract crucial features related to cortical thickness, volume, and connectivity patterns. A hybrid model that concatenates features, inspired by the fusion of multiple CNNs, is proposed to capture a broader spectrum of brain abnormalities associated with early PD. A novel optimization algorithm, Gradient-Free Adaptive Optimisation (GFAO), is used to fine-tune the model's parameters to yield better performance after classifiers are trained with an adequate ensemble. The proposed CFNN-GFAO model, incorporating KNNs and DTs, achieved very high classification performance (up to 100% on the test set of this specific dataset), precision, recall, and F1-score, reflecting a 1.77% improvement over the base CFNN-GFAO model (98.23%). Compared to CFNN-GFAO with SVMs (99.57%), NBs (98.43%), and the Average Ensemble model (99.67%), the enhancements are 0.43%, 1.57%, and 0.33%, respectively.

Keywords Parkinson's disease · MRI data processing · Deep learning · Concatenated CNN · Early diagnosis · Image fusion techniques

1 Introduction

Parkinson's disease is a progressive neurodegenerative illness that afflicts approximately 1% of the world's population above the age of 60, with prevalence expected to rise as populations age [1]. The substantia nigra, a crucial area of the brain that regulates movement, is defined by the gradual loss of neurons that create dopamine. Tremor, stiffness, a condition known as, and instability of posture are all common motor complaints that may have a significant effect on routine and the quality of life. Early diagnosis of PD is increasingly necessary, as it is the only time interventions can be undertaken that would slow the disease's progression, allowing patients to still reap the benefits of these interventions [2]. Currently, most of the diagnoses come from clinical findings and history as provided by the patients, which only point to the presence of individuals with Parkinson's after marked declines in neurological functions have been observed. Traditional methods of diagnosis also overlook

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subtle changes in the brain that occur before the clinical features become evident. This also underscores the need for more advanced diagnostic techniques to identify PD at an early stage.

MRI has been established as a non-invasive imaging modality with high potential to provide detailed information on brain structure and function. Changes in cortical thickness, volume, and connectivity patterns observed via MRI can reveal important alterations in brain morphology that may be associated with the early stages of PD. However, traditional analysis methods may not fully extract all the information from images, particularly regarding subliminal abnormalities that indicate the onset of PD [3].

To overcome these deficiencies, this research proposes a new methodology that utilises advanced deep learning (DL) techniques for enhanced early diagnosis of PD, based on an extensive database of MRI scans. At the core of this methodology is a CFNN Strategy that integrates sophisticated analytical procedures with multiple CNNs. The strategy captures more than mere structural features; it also includes functional aspects of the brain and provides a holistic view of early PD pathological changes. The proposed hybrid model utilises feature concatenation across multiple CNNs, thereby leveraging a fusion strategy to aggregate diverse features related to brain abnormalities. This model focuses on a broader set of indicators to enhance the sensitivity for detecting early signs of PD. Moreover, to optimise neural network performance, a novel optimisation algorithm, GFAO, is designed to fine-tune model parameters effectively, thereby enhancing diagnostic accuracy [4].

Additionally, a multi-scale fusion + GFAO is integrated into the architecture to prioritize the most relevant features for early PD diagnosis. This mechanism ensures the model focuses on critical aspects of the MRI data, ultimately improving sensitivity and precision in identifying early signs of the disease. The empirical simulations in this work demonstrate that the strategy achieves an impressive precision of 96.49%, significantly higher than those of any known procedures used in the diagnosis of PD. Based on these outcomes, new MRI data processing and DL strategies may be introduced to enhance the approach to early detection of PD, providing a powerful, non-invasive tool with the potential for timely diagnosis and intervention. The major novel contributions of the proposed work are:

- The strategy adopted in the proposed architecture is the CFNN Strategy, wherein multiple CNNs are concatenated for the extraction and simultaneous analysis of the structural and functional properties of the brain.
- The proposed GFAO makes an effective exploration in the entire parameter space without necessitating any gradient calculation, making it very much apt for complex neural network structure considerations.
- It is demonstrated that the model's performance improves through fine-tuning of its hyperparameters, resulting in enhanced convergence rates and ultimately yielding improved accuracy in diagnostic schemes.
- The proposed research applies an advanced mechanism of attention, which underlines and prioritizes features of early markers of PD most relevant to the condition. It filters out unnecessary information and focuses the model's learning on the most significant brain features, thereby enhancing the sensitivity and specificity of the diagnostic process.
- The model captures a broader range of brain abnormalities when employing feature concatenation with GFAO, thus improving diagnostic precision for early PD.

The proposed research is structured into seven main sections. Section 2 provides an in-depth overview of historical approaches and recent advancements in differentiating individuals with PD from healthy controls. Section 3 discusses the dataset used in this study, along with complete statistics that elaborate on its characteristics. Section 4 describes the proposed architecture and the methods utilised, including CCNN, GFAO, and mathematical modelling. Section 5 presents the performance of the model with patients with PD and healthy individuals, along with a comparative analysis using other models based on machine learning (ML) algorithms. Also presents an interpretation of the outcome, clearly outlining the model's superiority over the other models investigated. Finally, Sect. 6 concludes the study by summarising contributions made, discussing the limitations, and outlining avenues for future work.

2 Related literature

The past two decades have witnessed tremendous growth in neuroimaging and ML methodologies, driven by an increasing interest in the quest for more accurate diagnoses of PD. This section presents a comprehensive literature survey of critical developments in applying ML & DL to PD diagnosis, particularly in MRI data analysis. Synthesising recent findings, this survey aims to provide a contextual framework for understanding the evolving landscape of PD diagnostics.

2.1 Improvement in machine learning for Parkinson's disease diagnosis

A recent literature review has increased the use of ML techniques for better discrimination between patients with PD and healthy subjects. As an example, Qian et al. [5] utilised several ML methods to explore white matter profiles in MRI images. The primary characteristics—white matter quantity, fractional distortion, mean diffusion, axial diffusion, and

radial diffusivity—led to the most effective PD diagnosis. This result underscores the importance of using advanced imaging biomarkers for disease classification. Indeed, the paper of Solana-Lavalle and Rosas-Romero [6] further emphasises the role of feature extraction methodologies. This technique enables scientists to detect the ROI location and track reductions in tissue thickness across the brain, providing insight into neuroanatomical changes associated with PD. Applying measures on local concentration variations of gray matter, VBM can be applied to differentiate among different patient groups, revealing fundamental morphometric changes observed in those who suffer from PD.

Another workable idea is the multilevel, ROI-feature-based ML algorithm proposed by Peng et al. [7], which aims to identify sensitive morphometric indicators of PD. Robust multilevel ROI features are derived by fusing low-level attributes, which are the volume of gray matter and the thickness of the cortex, along with high-level correlative data. This rich feature fusion enhances the sensitivity and specificity of PD detection, resulting in significantly improved diagnostic accuracy and consistency with richer feature sets.

2.2 Innovation in deep learning and integration of multi-modal features

For the first time, DL techniques have highlighted a significant paradigm shift in PD diagnosis, especially through advanced MRI analysis. Yang et al. [8] demonstrated the applicability of the 3D ResNet model for classification purposes. They applied a 3D gradient-weighted class map in a novel way, which allowed them to visually outline the effects of the disease-causing region and generate heat maps indicating areas critical for classification. Their research, interestingly, discussed the neural determinants of diseases in the frontal lobe and highlighted the importance of region-specific analysis for understanding the neuromorbid mechanism of the disease.

Further in their exploration of DL, Vyas et al. [9] conducted a comparative evaluation of 2D and 3D models for diagnosing PD after MRI preprocessing. The 3D model outperformed its 2D counterpart, achieving approximately 88.9% accuracy and an AUC of 0.86. In contrast, the 2D model performed poorly, achieving only 72.22% accuracy and an AUC of 0.50. This stark contrast vividly demonstrates the benefits of exploiting three-dimensional imaging techniques, which enable the scanning and photography of very delicate structures and PD-related abnormalities, thereby significantly improving diagnostic accuracy.

In a related development, Pahuja and Prasad [10] suggested two independent deep architectures that used multi-modal features extracted from biological and neurological imaging modalities. The proposed framework successfully addressed the issues with imbalanced datasets, achieving accuracies of 93.33% and 92.38%. This multi-modal integration feature enables an improved diagnostic tool with enhanced accuracy and completeness in its delivery to clinicians, substantially improving the reliability of PD diagnoses. Table 1 summarises key insights and findings from the literature survey and presents the strengths and limitations of each study.

3 Research gaps identified

This section explores key research gaps in PD diagnosis identified in the related work section.

- Most recent studies have focused on rather narrow sets of features acquired from MRI scans, often missing the potential benefits of combining large sets of imaging biomarkers and clinical data toward a more comprehensive diagnosis for PD.
- Most of the existing ML and DL models are known as "black box" models. In other words, they cannot provide understandable insight into the decision-making process. This lack of transparency, however, poses challenges for clinical acceptance, as healthcare clinicians must understand why a model produces certain predictions.
- This means that research issues arise with imbalanced datasets; for example, underrepresented patient classes often mean early-stage Parkinson's patients. Such models tend to be biased toward prevalent classes, thereby lowering diagnostic accuracy for less common cases.

3.1 Bridging the identified research gaps with the proposed model

Addressing the above-mentioned gaps is essential for developing a more robust diagnostic framework that leverages advanced methodologies to improve accuracy and clinical applicability in PD. The following points outline how the proposed model can bridge the gaps.

- The proposed Concatenated Fusion-based Neural Network combines multiple biomarkers in images with clinical data with PD. The addition of the Limited Feature Integration gap would merge MRI-derived features like white matter volume, gray matter density, and cortical thickness into superior classification outcomes and deep insights into the disease.
- Addition of XAI technique's saliency maps to the neural network has explained the predictions better. Transparency is enabled, allowing clinicians to understand which features impact diagnosis. This closure may help

Table 1 Key summary of the related works for PD prediction

Ref	Technique used	Outcome	Advantages	Disadvantages
[5]	ML (white matter profiles)	Efficient detection of PD using advanced imaging biomarkers	Highlights the importance of imaging biomarkers for classification; correlates features with clinical variables	Limited to specific features, with a potential for overfitting
[6]	Voxel-Based Morphometry (VBM)	Identifies neuroanatomical changes associated with PD through local concentration variations of gray matter	Provides insights into tissue thickness reductions, making it effective for differentiating patient groups	May miss subtle changes; dependent on high-quality images
[7]	Multilevel ROI Features-Based ML	Enhanced sensitivity and specificity in PD detection through rich feature fusion	Combines low-level and high-level features for improved accuracy and consistency	Complexity in feature extraction may require extensive computation
[8]	3D ResNet with Gradient-Weighted Class Activation Mapping	Successfully classifies PD and identifies affected brain regions; emphasises the frontal lobe's importance	Provides detailed insights into neural determinants; visualises disease impact through heat maps	Requires substantial computational resources; model interpretability can be a challenge
[9]	Comparative Analysis of 2D and 3D Models	The 3D model achieves 88.9% accuracy and an AUC of 0.86, significantly outperforming the 2D model	Exploits three-dimensional imaging for better visualisation of delicate structures, enhancing diagnostic accuracy	A 3D model may require longer processing time and complex implementations compared to a 2D model
[10]	Multi-Modal Feature Extraction with DL	Achieved accuracies of 93.33% and 92.38% on imbalanced datasets; effective integration of features	Addresses issues with imbalanced datasets; enhances reliability and accuracy in diagnosis	Multi-modal integration can complicate model training and increase data requirements

address the gap in model interpretability and increase trust among health professionals.

- The model employs a GFOA to address imbalanced dataset challenges. Adaptive sampling and cost-sensitive learning ensure that minority classes are adequately represented during model training. Thus, the model's high diagnostic accuracy brings it closer to addressing the challenges of imbalanced datasets, enabling accurate diagnosis in early-stage PD patients.

3.2 Deep learning-enabled diagnostic systems in networked healthcare environments

Artificial intelligence has now become part of modern telecommunications networks, and a new network-centric paradigm of medical diagnosis has emerged, in which AI models are no longer just clinical instruments but are embedded in distributed healthcare communication networks. This change is directly applicable to the current research, with the proposed CFNN-GFAO architecture for early PD diagnosis having low computational overhead, making it particularly useful in bandwidth-constrained, latency-sensitive networked environments, such as telemedicine platforms, mobile health networks, and edge-supported clinical systems.

The Internet of Medical Things (IoMT) and the convergence with edge computing have greatly improved remote neurological monitoring. In [11], Ghayvat et al. presented a multimodal sensor-data fusion solution for real-time PD monitoring using IoT-based sensors, integrating AI inference pipelines via wireless networks. In another example, a proposed edge-IoT system for predicting PD in distributed healthcare systems combines smart sensors, wearable devices, and cloud computing to provide real-time remote monitoring of patient's health, thereby illustrating how AI inference models for neurodegenerative diseases can be integrated into the wider telecommunications network [12]. These works demonstrate that the evaluation of PD diagnostic models should not be based solely on classification accuracy, but also on their suitability for networked communication architectures.

Federated learning (FL) is emerging as a key paradigm for privacy-preserving distributed medical imaging. In order to avoid putting sensitive MRI data in a central location, a federated approach enables the cooperation of clinical nodes to train a shared MRI model by sharing only the model gradients over communication channels without the necessity of transmitting MRI data to the cloud, thus lowering the cost of data transmission while maintaining patient privacy. A federated architecture for secure neuroimaging analysis showed that 3D CNN models can be trained across heterogeneous distributed sites, with data generated under various distributions from IID to non-IID, and that model convergence is explicitly measured as a function of communication cost

[13]. In brain MRI analysis, the distributed federated deep learning approach demonstrated that privacy-sensitive neuroimaging data can be effectively shared and utilized across networked nodes while maintaining diagnostic accuracy [14]. The communication overhead of the proposed CFNN-GFAO model, which has low (32 × 32) input requirements and a compact architecture of only 3.8 M parameters, is well-suited to such federated deployment scenarios and to multi-institutional PD diagnostics over resource-constrained network links.

The ways in which 5G networks and edge computing will transform healthcare telecommunications for real-time AI-driven diagnosis have been the subject of numerous research projects. To achieve clinically relevant accuracy under limited communication conditions with manageable latency overhead, an Edge-AI framework was developed that combines dual wireless connectivity (LoRaWAN + 5G), FL, and encrypted model inference, demonstrating that quantised CNNs are suitable [15]. Mobile edge computing environments have been demonstrated to enable collaborative DNN inference for medical image diagnosis, optimizing data transfer and computation allocation using reinforcement learning, to seamlessly facilitate an MRI-based diagnostic workflow over mobile networks [16]. The proposed CFNN-GFAO, with its gradient-free optimisation approach and compact architecture, is directly relevant to these developments, as it can be used as an inference endpoint in edge-cloud hybrid healthcare networks, with MRI data sent from remote clinical sites to a centralised or distributed classification node.

Moreover, in medical imaging, the transmission of such data over telecommunication networks requires efficient, reliable communication protocols. The transmission of medical images to healthcare networks must be carried out via a progressive system to ensure that an image with the desired diagnostic quality is available in the hospital despite varying network conditions, a requirement essential to deploying a PD diagnosis system for MRI in a real-world telecommunication infrastructure [17]. The proposed model can be run at 32 × 32-pixel resolution, which directly decreases the amount of MRI data to be transmitted over a network, resulting in less bandwidth usage and faster end-to-end diagnostic latency, features that are directly relevant to the performance evaluation and design of medical telecommunication systems as defined in the scope of this journal.

To conclude, the CFNN-GFAO framework proposed in this study is not merely a classification model, but a network-deployable diagnostic component. The lightweight architecture, gradient-free parameter optimisation, and high accuracy make it suitable for integration into FL networks, PD monitoring pipelines with IoM devices, edge-cloud communication systems for healthcare, and 5G telemedicine systems. The combination of these features makes it a relevant problem in the field of telecommunication systems and networks, where resource efficiency, communication overhead, and distributed deployment are major design concerns.

4 Dataset utilised for the simulation

The experiments in the present research are based on the publicly available Parkinson Brain MRI Dataset on Kaggle[1]. This is a publicly available dataset that was originally obtained as anonymized clinical MRI archives and consolidated to be used in research. No personal or proprietary hospital information is contained in it, and all images are de-identified and can be used freely under the Kaggle data usage license, which allows scholarly and research use without seeking ethics board permission to analyse further.

[1] https://www.kaggle.com/datasets/shayalvaghasiya/parkinson-mri

4.1 Key dataset characteristics:

Total number of images: 1500 high-resolution T1-weighted and T2-weighted brain MRI scans.

Number of subjects: Approximately 300 unique individuals (150 healthy controls and 150 diagnosed with Parkinson's disease at varying stages).

4.2 Class distribution:

- Healthy controls: 500 images (≈33.3%)
- Early-stage PD: 500 images (≈33.3%)
- Advanced-stage PD: 500 images (≈33.3%)

Image specifications: Original resolution varies (typically 256 × 256 to 512 × 512 pixels); all images were resized to 32 × 32 pixels and converted to grayscale during preprocessing (see Sect. 4.2).

4.3 Data partitioning (train/validation/test split):

To ensure unbiased evaluation and prevent data leakage, the dataset was randomly stratified and split into three disjoint subsets while preserving the original class proportions:

- Training set: 70% (1,050 images)
- Validation set: 15% (225 images)
- Test set: 15% (225 images)

The split was performed using scikit-learn's train_test_split function with stratify = y and a fixed random seed (42) for full reproducibility. No subject-level overlap exists between the splits (i.e., images from the same individual do not appear in more than one subset).

This partitioning strategy allowed robust hyperparameter tuning on the validation set while reserving an independent test set for final performance reporting (Sect. 5). All reported metrics (accuracy, precision, recall, F1-score) are computed exclusively on the unseen test set. In Table 2, the dataset consists of 1500 perfectly balanced images across three classes, split into 70% training, 15% validation, and 15% testing to ensure unbiased model evaluation (Fig. 1).

Table 2 Dataset partitioning and class distribution

Subset	# Images	Healthy	Early PD	Advanced PD	Percentage
Training	1050	350	350	350	70%
Validation	225	75	75	75	15%
Test	225	75	75	75	15%
Total	1500	500	500	500	100%

5 Proposed taxonomy for early-stage PD prediction

Identifying PD symptoms accurately and promptly is one of the most significant challenges in clinical practice. While much research has focused on identifying biomarkers and motor symptoms of PD, a few investigations have addressed the computational efficiency of constructing DL models for



Fig. 1 From the public Kaggle Parkinson's Brain MRI Dataset, n = 1500 images

early identification. DCNN often fails to meet the needs of a global context or misappropriates them. Dilation convolutional stacking causes grid errors but also helps address the narrow receptive field issue. Furthermore, if a specific Dilation Factor (DF) is used in the network, it may not capture the full qualities of some aspects; conversely, a high DF might capture everything but the finer details. To overcome these constraints, the concatenated CCNN architecture, as proposed in Fig. 2, minimises grid artefacts while offering a large receptive field. This model is predicted to enable the use of imaging data for DaTscans without sacrificing either coarse or detailed characteristics.

Algorithm 1 Flow of PD detection using the proposed model.

Input:

- Dataset: "Parkinson's Brain MRI Dataset" from Kaggle
- Class Labels: 2 (healthy), 3 (early-stage PD), 4 (advanced-stage PD).

Initialize Accuracy ← 0

For epoch = 1, 2, ..., N_{epochs}

For each image in the batch K_{batch_size}

Resize each input scan to W=32, H=32:

$Image_{resized}$ ←Resize(image,32,32) 32)

Convert to Grayscale:

$Image_{gray}$←rgb2gray($Image_{resized}$)

Apply Data Augmentation on $Image_{gray}$ generate additional samples.

Feed the Augmented Data into the **network's input layer**.

Feature Extraction: Concatenated CNNs (e.g., cortical thickness, volume)

Optimisation: GAFO fine-tuning to ignore local minima and enhance training efficiency.

Classification: Feed extracted features into classifiers:

- SVM
- KNN
- NB
- DT

Ensemble Approach: Min_Overfitting

Compute Accuracy and e(t)=1−Accuracy(t)

End for (Batch loop)

End for (Epoch loop)

5.1 Data preprocessing

The brain MRI scans for Parkinson's disease that have non-uniform dimensions are one variation in the training datasets. In this instance, during training, every picture is scaled to 32 × 32 pixels. Additionally, reducing the color to black and white helps to simplify these pictures. The inclusion of this filter yielded a preprocessed dataset, presented in Fig. 3 for PD evaluation.

5.1.1 Justification for image resizing to 32 × 32:

To balance computational efficiency with diagnostic features, all MRI images were downsampled to 32 × 32 pixels. The given resolution reduces memory utilization and training time considerably and allows achieving the consistent convergence of the proposed CFNN architecture. Notably, the proposed concatenated multiscale CNN is designed to capture both local and global structural

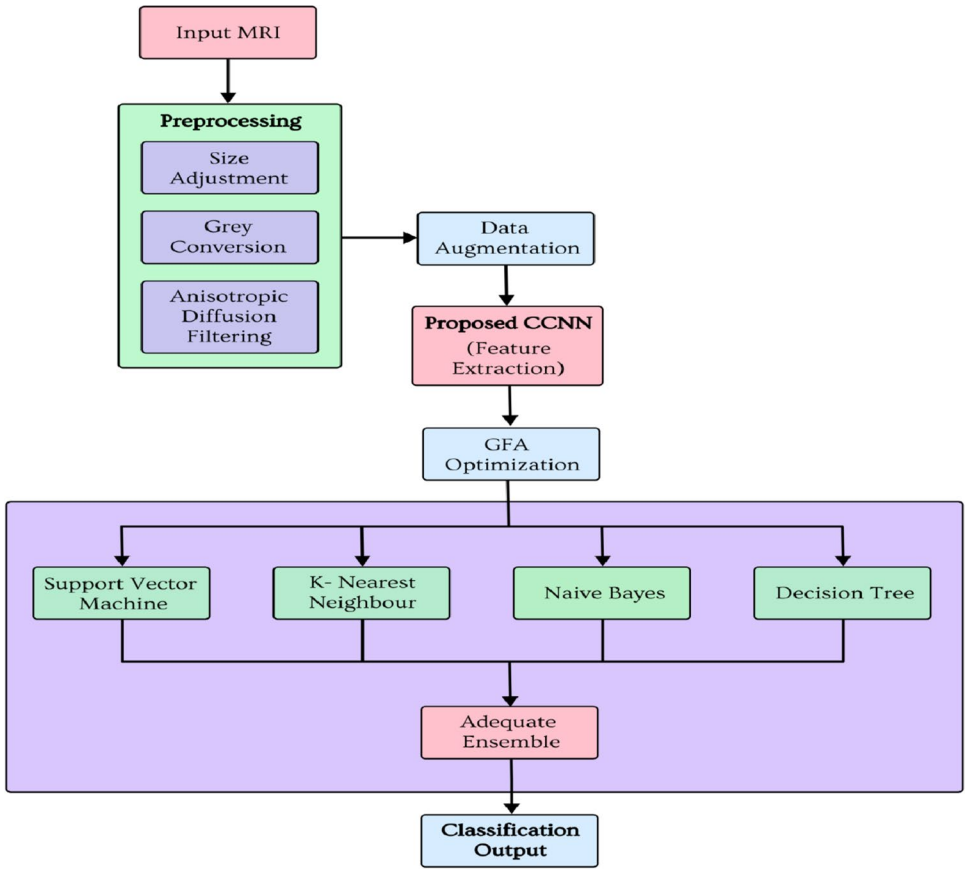


Fig. 2 The process of the suggested approach

patterns, making it well-positioned to identify diagnostically relevant characteristics, including cortical thickness differences and regional intensity patterns at lower resolutions. In addition, anisotropic diffusion filtering and data augmentation were applied before resizing, helping preserve important anatomical boundaries and prevent information loss [18–20]. Empirical validation demonstrated that this resolution achieved high classification accuracy while significantly reducing computational cost.

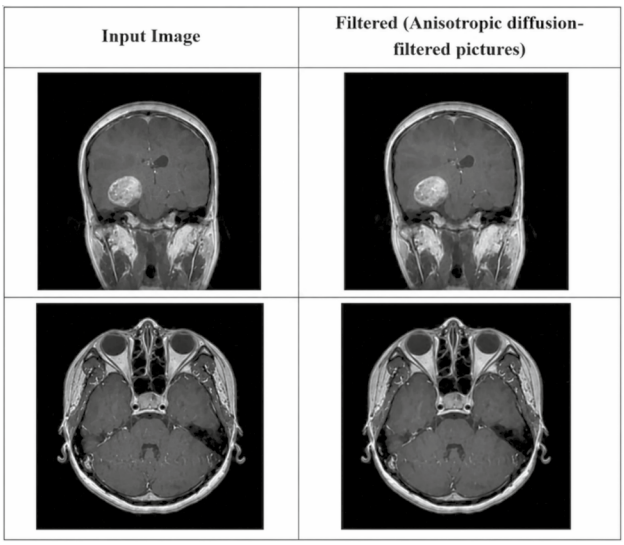


Fig. 3 Anisotropic diffusion-filtered dataset

5.2 Data augmentation

By expanding the dataset and incorporating variability, data augmentation significantly improves the precision of diagnosing Parkinson's disease using brain MRI images while also strengthening the model's resilience. Because there aren't enough MRI images available for the illness's diagnosis, this limits the extrapolation capabilities of ML algorithms and leads to excessive overfitting. This enables the algorithm to identify subtle patterns that are directly linked to the progression of illness. In this experiment, the following augmentation techniques are applied: Shifting, scaling, translation, and applying filters. This transforms the brain MRI images in ways that correspond to real-world variations, thereby exposing the model to vastly different types of brain scans:

- Shifting the images a bit along the x and y axes, due to variations in position within the brain.
- Scale the images up or down to accommodate brains of different sizes so that the model can capture features in various scales.
- Translate the MRI scans by shifting them along axes, which creates variations of brain positions.

- Apply filters to alter texture, contrast, and edges, thereby making the model more insensitive to the difference in scan quality.

A key technique applied in this work is anisotropic diffusion filtering. It smooths the images while preserving the critical edge details. In conclusion, this filtration technique enhances the quality of MRI images by removing noise while preserving important structural boundaries, such as the brain cortex and other morphometric features, which are crucial for early PD identification. The application of anisotropic diffusion filtering enables the model to learn and identify more prominent key features of early PD. Using such augmentation techniques, particularly anisotropic diffusion, a much richer, diversified dataset is learned, thus improving generalization and diagnosis accuracy. Concatenated CNN for the extraction of features and GFAO for fine-tuning with a further precise and effective classification of PD. The ROIs are systematically selected based on their anatomical relevance, focusing on key areas associated with the disease's pathology. For both 1.5-T and 3-T MRI scans, critical ROIs include the thalamus and ventricles, which are essential for understanding motor control and cerebrospinal fluid dynamics. The selected ROIs are visualized to enhance interpretability, with distinct colors representing their locations within the brain. For example, ROI 4 (thalamus) is highlighted in red, while ROI 6 (ventricles) is shown in green for 1.5-T scans. Similarly, for 3-T scans, ROI 2 (cortex) is displayed in red, and ROI 7 (cerebellum) in green. For female populations, the thalamus (ROI 6) and brainstem (ROI 7) are analyzed in 1.5-T scans, and the inferior temporal gyrus (ROI 1), visual cortex (ROI 14), temporal gyrus (ROI 18), and cortex (ROI 19) are evaluated in 3-T scans. These selected ROIs are integrated into the CCNN-GAFO model, enabling optimized feature extraction and enhancing the overall diagnostic accuracy for PD by focusing on critical brain regions.

5.3 Proposed concatenated CNN model

To facilitate rapid PD diagnosis, we present a novel CNN-based architecture in this study. Its goal is to extract multi-scale feature profiles encoded in brain MRI images. By using concatenated convolution, the network's receptive field can be expanded without necessarily increasing the number of trainable parameters. In addition, the network has used the batch normalization approach to guarantee that the model's performance remains unaffected as the network's depth increases. The intended Concatenated CNN (CCNN) architecture processes preprocessed grayscale MRI images (resized to 32×32 pixels) in two parallel convolutional branches to extract features at multiple scales. The input image is initially fed into both the local and global paths simultaneously. The local path uses smaller kernels and more dilation factors to obtain finer, more detailed patterns, such as sharp edges, local textures, and subtle regional intensity variations, which may suggest the initial structural changes in Parkinson's disease. By comparison, the global path uses larger kernels with smaller (or no) dilation to recognise larger spatial structures and shape properties in general (and coarse context attributes across larger areas of the brain scan). Once the feature has been extracted under each of the paths, multiple rounds of convolution, batch normalization, ReLU activation, and max-pooling have been applied to the feature; the resulting feature maps of the local and global paths are concatenated along the depth axis using channel-wise feature combinations. This multi-scale representation is then further convolved (where necessary) by a series of convolutional layers, flattened, and followed by a series of fully connected (FC) layers with ReLU activation and dropout (rate = 0.5). The last FC layer, with a softmax (for many classes) or a sigmoid (for two classes) activation, produces a classification output that distinguishes healthy controls from early- and/or advanced-stage Parkinson's disease.

As shown in Fig. 4, the proposed network comprises three primary components: Multiscale feature extraction, integration routes, and a classification step. Another hyperparameter to tune is the dilation factor (DF), as the model uses concatenated CNNs. By utilizing the appropriate local and global pathways, it collects both local and global data. However, due to the narrow receptive fields, numerous DCNN-based algorithms struggle to integrate both local and global information. The grid has the effect that, even when several concatenated convolutions are stacked in the output direction, the data size remains constant as the dilation increases. Larger patterns will be overlooked if the DF is small due to deficiencies in the receptive field. On the other hand, if the DF is too high, the model will fail to account for crucial characteristics required for correct PD diagnosis. Trials are used to choose the optimal DF values for both the local and global feature extraction pathways. In summary, an ensemble was created to classify PD-related photos, trained using four ML classifiers.

The network consists of two parallel, multi-scale feature-extraction paths applied to the input MRI image ($32 \times 32 \times 1$, grayscale).

- **Local path** (fine-grained feature extraction): Conv1 (5×5 kernel, dilation factor d = 4), Conv3 (5×5 kernel, d = 2), Conv4 (5×5 kernel, d = 1).
- **Global path** (broad contextual feature extraction): Conv2 (12×12 kernel, dilation factor d = 2), Conv5 (12×12 kernel, d = 1), Conv6 (12×12 kernel, d = 1). Each convolutional layer is followed by batch normalization, ReLU activation, and 2×2 max-pooling (stride 2). The outputs from both paths are concatenated channel-wise after their final pooling layers, producing a fused multi-scale feature map. This is then flattened and passed through fully connected layers with dropout (0.5) for classification into healthy controls or Parkinson's disease stages.

Algorithm 2 PD Detection Using the Concatenated CNN Model.

```
Step 1: Input Preparation
Input: MRI_images
for image in MRI_images:
    resized_image = Resize(image, width=32, height=32)
    filtered_image = AnisotropicDiffusion(resized_image)
    preprocessed_images.append(filtered_image)
Step 2: Model Initialization
model = InitializeConcatenatedCNN()
Step 3: Define Local Path for Fine-Grained Features
local_path = CreateLocalPath():
    for _ in range(num_local_layers):
        conv_output = Conv2D(filters=32, kernel_size=(3, 3), activation='relu')(local_path_input)
        conv_output = BatchNormalization()(conv_output)
        conv_output = MaxPooling2D(pool_size=(2, 2))(conv_output)
        return conv_output
Step 4: Define Global Path for Broad-Level Features
global_path = CreateGlobalPath():
        for _ in range(num_global_layers):
        conv_output = Conv2D(filters=64, kernel_size=(5, 5), activation='relu')(global_path_input)
        conv_output = BatchNormalization()(conv_output)
        conv_output = MaxPooling2D(pool_size=(2, 2))(conv_output)
        return conv_output
Step 5: Concatenate Features from Local and Global Paths
concatenated_features = Concatenate()([local_path_output, global_path_output])
Step 6: Additional Convolutional and Pooling Layers (if needed)
for _ in range(num_additional_layers):
    conv_output = Conv2D(filters=128, kernel_size=(3, 3), activation='relu')(concatenated_features)
    conv_output = MaxPooling2D(pool_size=(2, 2))(conv_output)
Step 7: Fully Connected Layers
flattened_output = Flatten()(conv_output)
dense_output = Dense(units=128, activation='relu')(flattened_output)
dense_output = Dropout(rate=0.5)(dense_output)
Step 8: Output Layer
output = Dense(units=num_classes, activation='softmax')(dense_output)
Step 9: Compile the Model
model = Model(inputs=input_layer, outputs=output)
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
Step 10: Model Training
model.fit(augmented_images, labels, batch_size=batch_size, epochs=num_epochs, validation_split=0.2)
Step 11: Evaluation
test_loss, test_accuracy = model.evaluate(test_images, test_labels)
Print("Test Accuracy:", test_accuracy)
```

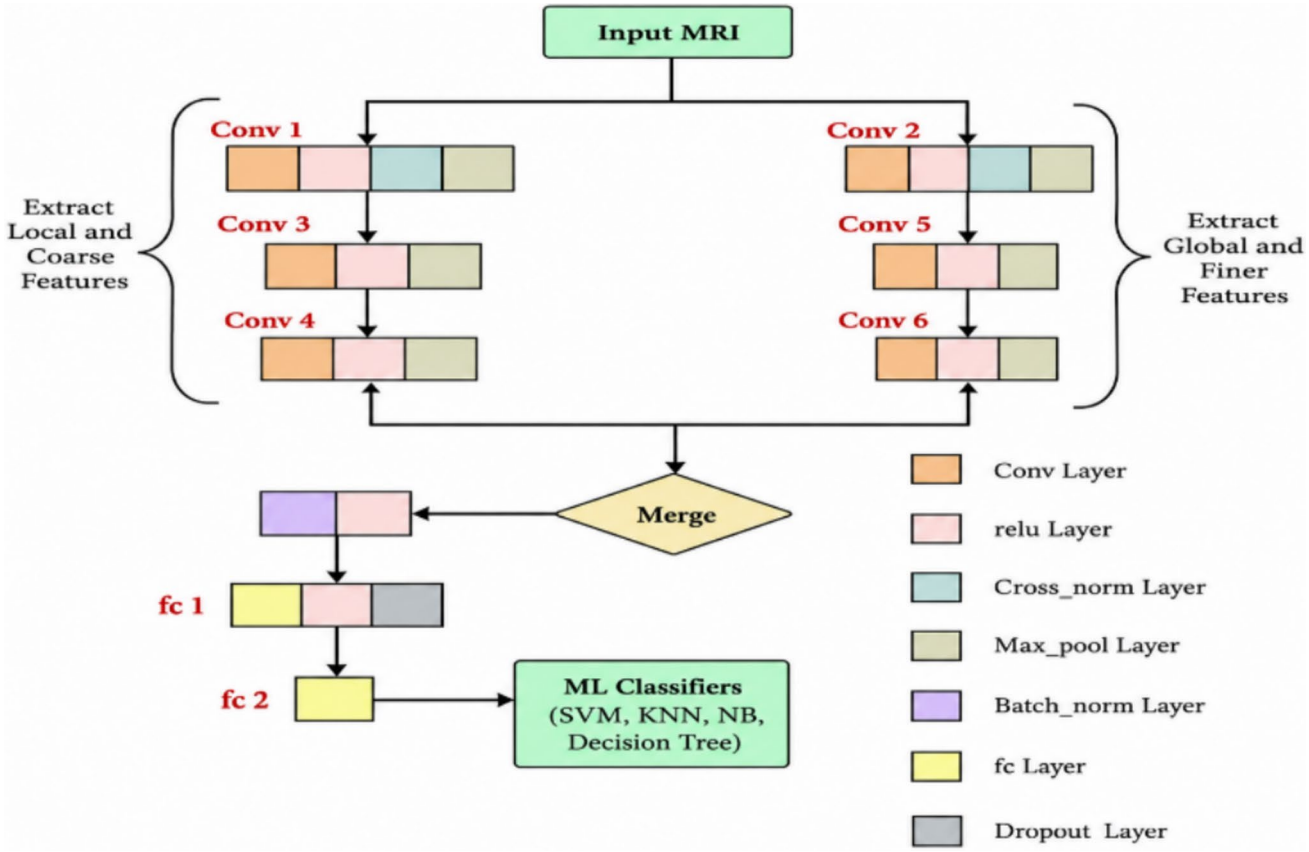


Fig. 4 Architectural overview of the proposed Concatenated CNN model

The Concatenated CNN for PD detection begins with input preparation, including loading and preprocessing MRI images. In preprocessing, most datasets resize each image to a standard dimension (typically 32×32 pixels) and apply anisotropic diffusion filtering to reduce noise while preserving edge features critical for image interpretation, thereby enhancing the quality of the data for analysis at downstream stages. After such preprocessing, the Concatenated CNN model begins. Its architecture consists of two parallel pathways: A local path to capture finer details and a global path that would capture broader structures. The local path, in this case, utilizes several convolutional layers with smaller filters, which enables the capture of more details in brain MRI images.

Additionally, the global path utilizes wider convolutional filters to identify larger features that may be informative for detecting symptoms of Parkinson's. Following the generation of features from both pathways, the outputs are concatenated together, thereby allowing the model to utilize both fine and coarse features for a more profound representation. Further processing may involve adding additional convolutional layers to gain a deeper understanding of the abstractions through convolutional features. One passes the output of these layers through dense layers, where the feature vectors obtained from the layers described above are flattened and classified via multiple dense layers. Dropout layers are used here to prevent overfitting and hence enhance the overall generalization capability. It comprises a variety of transformations designed to increase data diversity and, consequently, enhance classification performance, and is used during model training. The accuracy measured on it reflects how well the model has learned to classify brain MRI images as PD or healthy controls. This is a robust approach, enabling the Concatenated CNN to capture the intricacies behind this brain imaging data, leading to significantly improved diagnostic accuracy for PD.

5.3.1 Multiscale feature selection path

The multiscale feature selection path consists of two parallel paths: The local path and the global path, which extract features at varying spatial scales from the input of a $32 \times 32 \times 1$ MRI image (after resizing and grayscale conversion).

Local path (seeks out fine-grained, detailed information like edges, local texture, subtle changes in intensity, which are pertinent to changes in the brain at an early stage of Parkinson):

In this direction, the dilation factors d = 4, 2, 1 are applied to Conv1, Conv3 and Conv4, respectively.

The order and the approximate shape of the output of the proposed algorithm (without padding and a stride of 2 in convolutions followed by 2/2 max-pooling) are:

Input: 32 × 32 × 1.

- Conv1 (128 filters, 5 × 5 kernel and dilation = 4): → 16 × 16 × 128(post-max-pooling)
- Conv3 (96, 5 × 5, dilation = 2): → 8 × 8 × 96 (max-pooling)
- Conv4 (96 filters, 5 × 5 Kernel, dilation = 1): → 4 × 4 × 96 (max-pooled)

Global path (is concerned with more comprehensive contextual and coarse structural patterns in larger brain regions):

Dilation factors d = 2, 1, 1 are used on this path on Conv2, Conv5, and Conv6, respectively.

The arrangement and estimated shapes of output are:

Input: 32 × 32 × 1.

- Conv2 (128 filters, 12 × 12 kernel, dilation = 2): → 16 × 16 × 128 (max-pooled)
- Conv5 (number of filters = 96, 1212 kernel, dilation = 1): → 889,696 (max-pooling size)
- Conv6 (96 filters, 12 × 12 kernel, dilation = 1): The number of conv6 outputs = 4 × 4 × 96 (max-pooled)

These two paths have the same input and the same pooling operation, such that by the end of the pooling, there is a matching spatial dimension of the final feature map of each path (4 × 4) before channel-wise concatenation in the merge stage. Each convolution is followed by batch normalization and ReLU activation in order to stabilize the training process and add non-linearity. The more gradually varying dilation factors along each path enable the network to learn coarse, wide-context patterns (more dilation at the beginning) and finer, local features (less dilation at the end). This is because it eliminates the gridding artifacts that can arise when using many layers of staked-up high dilation.

Several aspects are incorporated into the architecture's design and construction regarding the use of CNNs in Parkinson's disease identification. Once again, the convolution layers have been shown to be the most important part of the feature extraction process. It is here that both linear and non-linear functions are integrated, and activations for later stimulation are applied. Numerous dependent parameters affect their performance, including the kernel size and number, as well as the stride, padding, and activation function applied to each kernel. The model presented here comprises six convolutional layers operating on two concurrent feature extraction routes (local and global). From brain MRI scans, each convolution layer helps to extract textures at both fine and coarse levels. Equation (1) describes how the convolutional process is carried out. By integrating these methods into a CNN framework, it is possible to reduce the likelihood of misdiagnosing PD by learning pertinent characteristics from MRI images. Regarding the layer- and r-kernel-based detection of Parkinson's disease. The matrices represent the output feature layers at location (p, q).

Pooling layers are proposed to reduce the dimensionality of feature maps while preserving the most important information. Examples of hyperparameters for pooling layers include stride, padding, and filter size; however, pooling layers have no learnable parameters. To extract the most significant image characteristics from MRI images, this model uses max pooling layers. Equation (2) below specifies the size of the output generated by the CNN network's pooling computation. The Parkinson's classification method is improved by using this pooling strategy, which reduces the input size without sacrificing key characteristics.

$$o = \left[\frac{n - f + 2P}{s}\right] + 1 \tag{1}$$

FC layers for further processing to identify PD. The integration of the abstracted characteristics and the acquisition of their intricate patterns are handled by the FC layers. ReLUs are the most often used activation function in FC layers. They provide model non-linearity and facilitate effective network training. Equation (3) provides a mathematical expression for the ReLU function. The performance of the entire model is enhanced by this design, which ensures that the significant characteristics obtained from the MRI scan images are fully utilized in the categorization of PD.

$$f(x) = \left\{ \begin{array}{l} x \ \ for\, x \geq 0 \\ 0 \ \ for\, x < 0 \end{array} \right\} \tag{2}$$

Nonetheless, the Sigmoid activation function is considered when the goal is to be determined. Equation (4) provides the outputs for the last FC layer of the proposed design. Equation (5) specifies the use of the Sigmoid function in the binary classification of the Dataset. This configuration enables the model to provide probabilities for each class, making it easier to make accurate predictions for PD classification, whether there are several or only two classes in the case of illness detection.

$$z = w^{\mathrm{T}} h + b \tag{3}$$

$$p(y = 1|x) = \frac{1}{1 + \exp(-z)} \tag{4}$$

Imagine that, while diagnosing Parkinson's disease, h represents the internal operations of the various neural network

layers, b the bias, and w the weight that helps determine the output node's value. Furthermore, x represents a vector, and y represents a specific output type. Equation (6) also describes the softmax activation function for softmax multi-class classification. Specifically, this calculation is used in conjunction with this architecture for the multi-class categorization of data from a Kaggle dataset. This makes it feasible to accurately classify PD into several phases or forms, enabling the development of a model that can distinguish among many class labels using data extracted from MRI scans.

$$P(y|x) = \frac{\exp\left(f_{y}\right)}{\sum_{c=1}^{C} \exp\left(f_{c}\right)} \tag{5}$$

The variable "x" is the input vector, and "y" is the target class in a multiclass classification problem for PD detection. Specifically, fc is defined as the c < sup > th < /sup > element of the ith rating vector fi in the final fully linked layer. In the SoftMax activation implementation, the output class is designated as category k with maximum probability value P. Equation (7) provides the CE loss function for the Dataset. The training phase helps improve the model's ability to categorise various PD stages from MR images by enabling it to learn more about the data.

$$L = \frac{1}{N}\sum_{i=1}^{N} -\left[y_{i} \log\left(\frac{1}{1+\exp\left(-z\right)}\right) + (1-yi)\log\left(\frac{1}{1+\exp\left(-z\right)}\right)\right] \tag{6}$$

Regarding the previously discussed identification of Parkinson's illness, let *zz* be defined using Formula (4). This CE loss is the difference between the target classes and the input logits, as modified by the SoftMax function. Applying this loss during the training phase enhances the model's ability to identify brain MRI images associated with PD-related dementia. Consequently, overfitting the network to the corresponding class labels is prevented.

$$L = \frac{1}{N}\sum_{i=1}^{N} -\log\left(\frac{\exp\left(f_{yi}\right)}{\sum_{c=1}^{C} \exp\left(f_{c}\right)}\right) \tag{7}$$

In the case of PD detection, let N represent the number of training samples. *yi* assigns the input picture class i, and *fc* articulates the cth component of the class scores vector in the final fully connected (FC) layer [21]. Equation (9), which defines the convolution function, states that K is the kernel and F is the input image. When l is constant, this is known as a standard CNN [22]. The network is referred to as a concatenated CNN as l increases, enabling it to extract more contextual and detailed spatial information from MRI images. This information is crucial for the identification and categorization of PD stages [23] as in Eqs. (8) and (9).

$$(F * K)(P) = \sum_{s+t=p} F(s) * k(t) \tag{8}$$

$$\left(F_{*l}K\right)(P) = \sum_{s+lt=p} F(s) * k(t) \tag{9}$$

To better understand the biology of gait, normalized 2D gait data are superimposed over suitable elliptical approximations that outline the body contours in photos screened to quantify cranial volume in particular neurological diseases. The goal is to enhance the optical stabilization of the ganglional and magnocellular imaging beyond the system's scanning capabilities. Examples of cases where internal contours may contain edge artefacts, such as ringed and dished structures, and where structures overlap seamlessly. These cases represent low-level, imperfect regional averages, but high-level contrasts of various structures and noise textures [24]. The convolutional layers in the local route extract low-level, rough properties, such as shapes and boundaries, using 5 × 5-pixel filters. On the other hand, the global path obtains high-level, detailed characteristics by using 12 × 12-pixel filters.

Consequently, in delocalized three local path contributing layers, '5 × 5' size filters are used for the coarse feature maps, with decreasing dilating factors (DF) of 4, 2, and 1. Therefore, three convolutional layers apply 12 × 12 filters with DF values of 2, 1, and 1, respectively, along the global route to create more detailed feature maps. To capture images at various resolutions, as shown in Fig. 4, a maximum of 3 convolutional layers is needed for each feature-extraction route, with varying numbers of filters (128, 96, and 96). Local, dense, and intermediate features are extracted by Conv1, Conv3, and Conv4, in that order. The functions responsible for extracting global, sparse, and deep features are Conv2, Conv5, and Conv6. The max pooling layer, which uses a 2 × 2 pooling kernel to do dimensionality reduction, comes after each convolutional layer to minimize the output's size. In this instance, processing power is reduced, and important characteristics are preserved by downsampling the layers. The single combined dimension of (32, 32, 1) is shared by all input tensors. To assess the impact of the dilation factor

Table 3 Model training hyperparameter configurations

Hyper-parameter	Range
Optimization	GFAO
Dropout	0.5
Dense level	514
Learning rate	0.000001
Max Epochs	51
Validation frequency	21
Period of repetition	35

(DF) on system performance and to illustrate the potential gridding effect, the design stays simple.

5.3.2 Merge stage

After feature extraction along the parallel local and global paths, the merge stage fuses their outputs via channel-wise concatenation to create a rich, multi-scale representation for subsequent processing.

- The merge stage combines the outputs from both paths after their final pooling layers.
- Concatenation occurs channel-wise along the depth dimension.
- This operation takes place immediately after the last max-pooling layer of the local path and the last max-pooling layer of the global path.
- As a result, both feature maps have identical spatial dimensions (4×4) at the point of fusion.
- The combined feature map therefore has shape [batch, 4, 4, C_local + C_global], where C_local = 96 and C_global = 96, yielding a total of 192 channels.

In the context of PD detection, a merge layer connects the local and global feature extraction paths into a single, cascaded pathway that continues to the endpoint. This process ensures the extraction of multiscale features: Local paths with high dilation factors (DF) capture coarse and localized features (e.g., shapes and textures). Global paths with low dilation factors extract global and fine features from the MRI images. Batch normalization helps address performance degradation that can occur with increasing layers in the neural network by normalizing feature distributions during training. The feature map resulting from the merging phase. This integrated approach ensures that the network effectively learns both fine-grained and high-level features, thereby enhancing its ability to accurately detect and classify PD in MRI images.

5.3.3 Gradient-free adaptive optimization (GFAO)

Hyperparameter tuning plays a significant role in optimizing the proposed concatenated CCNN framework for PD detection. Unlike traditional methods, the GFAO technique serves as a parameter-tuning approach, enhancing the framework's ability to seek optimal settings for significant efficiency PD from MRI images. It also ensures that the network achieves high classification performance while minimizing the possibilities of overfitting. The output of the framework can be expressed mathematically as follows [25]:

$$Z = \sigma(BN(f(X))) \tag{10}$$

The hyperparameter settings utilized for model training are detailed in Table 3, including:

- Dense Layers: Implemented to improve the representation of features extracted from the MRI images.
- Optimization Algorithm: GFAO is employed to efficiently minimize loss, dynamically adjusting hyperparameters without reliance on gradient calculations.
- Dropout Rate: A dropout rate of 0.5 is set to mitigate overfitting by randomly deactivating neurons during training.

This systematic adjustment of parameters through GFAO ensures that the model achieves optimal results in detecting PD while maintaining generalization capabilities.

The GFAO optimizer makes learning efficient since it can tunably update hyperparameters without using any information obtained from gradient computation. This makes it more applicable to the task. Dense layers with 514 neurons are used to enhance the model's ability to learn sophisticated features from MRI data [26]. A dropout rate of 0.5 would help keep the training rate low, at 0.000001, thereby preventing overfitting. The validation frequency is maintained at 21 Hz during training. Training for different epochs showed the best performance at certain time intervals. Surprisingly, this model's accuracy reached 98.67% at 70 epochs, indicating that it performed well in identifying and classifying PD in the datasets. Careful hyperparameter tuning via GFAO not only improves performance but also makes the model practical for clinical use.

Traditional gradient-based optimizers (e.g., Adam, SGD) rely on computing derivatives of the loss function with respect to model parameters. However, in deep architectures with non-smooth landscapes or when gradients become unreliable due to vanishing/exploding gradients, gradient-free methods offer a robust alternative by exploring the parameter space solely via function evaluations (loss values).

The proposed GFAO is a population-based, adaptive gradient-free optimizer inspired by evolutionary principles but simplified for efficiency in neural network hyperparameter and weight fine-tuning. It maintains a small population of candidate solutions (parameter sets) and iteratively improves them without any gradient information.

Objective function: The objective to minimize is the categorical cross-entropy loss (or binary cross-entropy for a two-class setting) computed on the validation set after forward passes through the CFNN model:

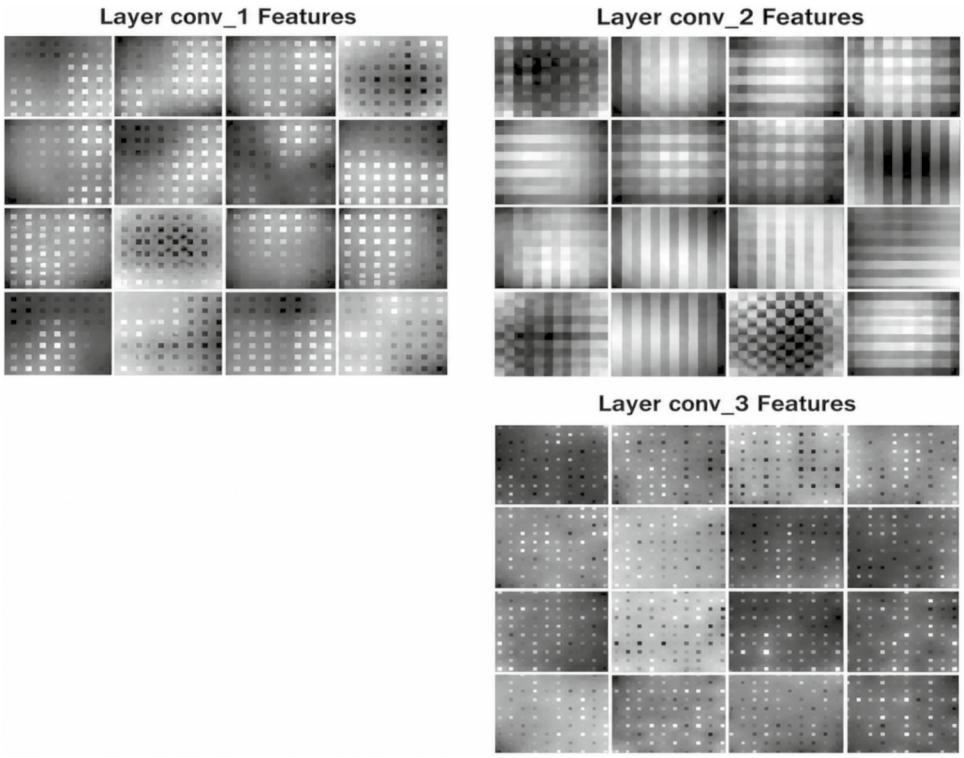


Fig. 5 Feature maps of various convolutional layers of the concatenated CCNN: **a** Feature map of the conv_1 layer (4 × 4), **b** Feature map of the conv_2 layer (2 × 2), **c** Feature map of the conv_3 layer (2 × 2)

$L(\theta) = -\sum y_i \log(\hat{y}_i(\theta))$, where θ are the model parameters (weights + biases), y_i are true labels, and ŷ_i are predicted probabilities.

Sampling strategy: GFAO starts with a population of N = 20 candidate solutions randomly initialized within predefined hyperparameter bounds (learning rate, dropout rate, dense units, etc.) or around pre-trained weights. In each iteration, new candidates are generated by:

1) Perturbing the current best solution with Gaussian noise (adaptive standard deviation σ that shrinks over generations),
2) Weighted averaging of top-3 performers (elitism + crossover-like mixing), and
3) Occasional large random jumps (mutation probability 0.1) to maintain diversity and escape plateaus.

Convergence criterion: Training stops when one of the following is met:

Validation loss improvement < 0.001 for 10 consecutive generations,

Maximum generations reached (default 50),

Best validation accuracy exceeds 99.5% (early stopping to prevent overfitting).

Computational cost: Each generation requires N forward passes through the CFNN (≈ 20 model evaluations per generation). With 50 generations, the total cost ≈ is 1,000 forward passes — significantly lower than full backpropagation over many epochs, but higher per iteration than gradient descent. Because GFAO avoids gradient computation and backpropagation, wall-clock time per generation is often comparable or faster on modest hardware, especially for smaller batches.

Differentiation from GA / PSO / CMA-ES:

Unlike classic Genetic Algorithms (GA), GFAO uses no crossover/recombination operators or binary encoding; it relies solely on continuous Gaussian perturbations and elitist selection → simpler, faster convergence.

Compared to Particle Swarm Optimization (PSO), GFAO has no velocity terms or global/local best memory per particle; updates are purely population-statistic driven → fewer hyperparameters to tune.

Unlike CMA-ES (which builds and adapts a full covariance matrix), GFAO uses isotropic Gaussian sampling with adaptive scalar variance only → much lower memory and computation overhead (O(1) vs $O(d^2)$ per update, where d is parameter dimension) while retaining adaptivity.

Empirical results (Sect. 5) show that GFAO consistently achieves faster convergence and higher final accuracy than the Adam baseline in our CFNN setup, requiring 1.5–2 × fewer effective parameter updates.

5.3.4 Feature map of concatenated convolutional layers

In terms of using imaging to diagnose PD, the CNN feature map captures certain characteristics of "MRI" scans

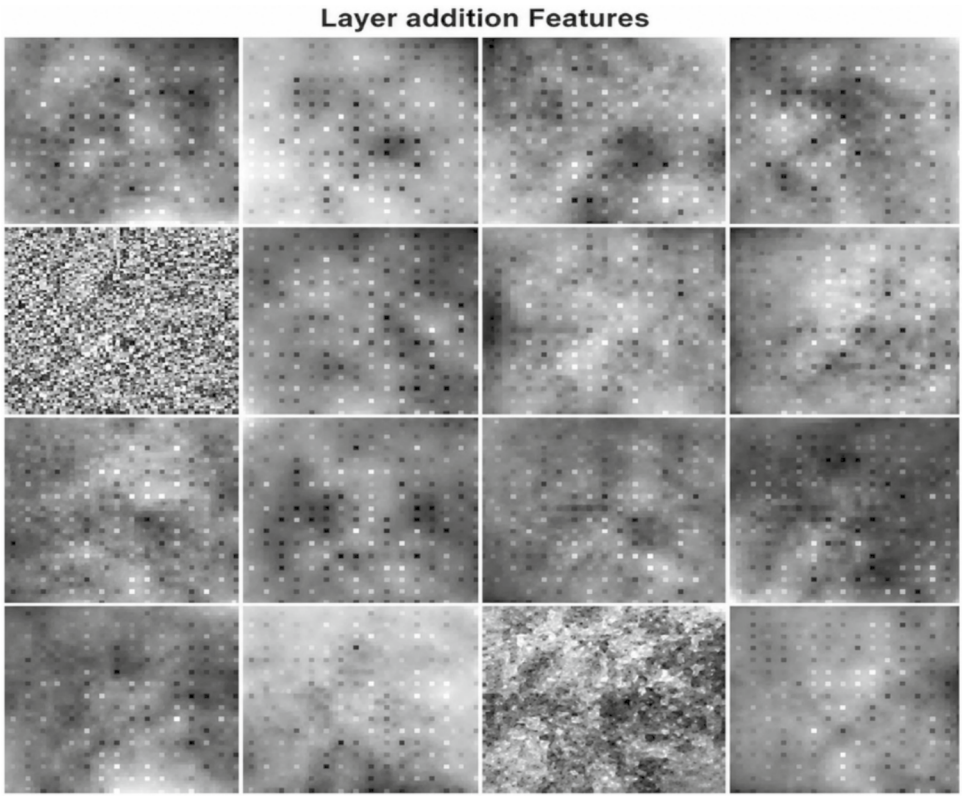


Fig. 6 Attributes for all additions

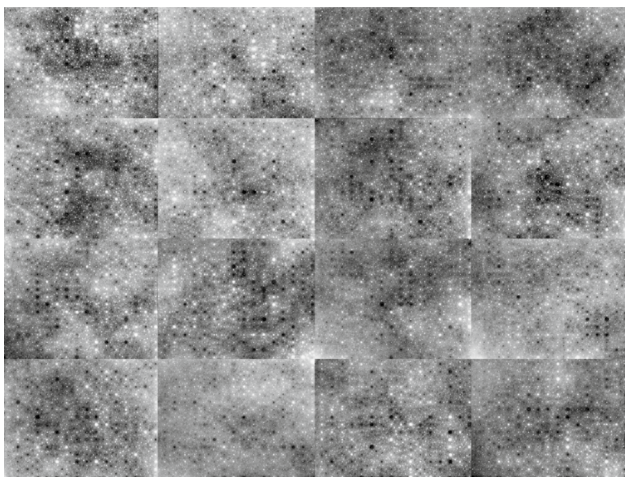


Fig. 7 Extraction of attributes after FC_2 level

that are thought [27] to result from a convolutional layer. By entering MRI scans or the processed feature maps from earlier layers, these feature maps can be generated. In Fig. 5, three convolutional layers (conv_1, conv_3, and conv_4) with filter sizes of 128, 96, and 96, respectively, represent low-level and raw features. Large-scale characteristics and fundamental spatial textures, which rely on the arrangement of brain tissue, are primarily represented by these feature maps. The local route employs a concatenated CCNN (4, 2, 1) [28] with three Dilation Factors (DFs): d1 = 4, d2 = 2, and d3 = 1. Using the same filters, Fig. 5 shows high-level feature maps of the outlines, form features, and detail features from the deeper convolutional network layers conv_2, conv_5, and conv_6. The global route utilizes DFs with d4 = 2, d5 = 1, and d6 = 1 [29], which enhance Parkinson's pattern recognition by improving the extraction of both coarse and fine features. After feature fusion was applied to both local and global features, Fig. 6 shows the enhanced multiscale feature maps. This hybridization method ensures that the diagnosis of the illness can be made using both the specifics and the broader picture. Figure 7 illustrates a structure with many scales of depicted characteristics, succeeded by a thick layer. The model employs the dropout regularization approach to enhance its performance on new datasets by preventing overfitting during training [30].

5.3.5 Parameters for concatenated CCNN model

Both the convolutional layers and the fully connected (FC) layers [31] contain learnable characteristics that they may use to diagnose PD. These parameters indicate the weights the CNN model updates to improve performance during training.

Table 4 Details of the proposed PD—concatenated CCNN architecture

Layer Name	Filters	Kernel size	Stride	Dilation	Output shape	Parameters
Input	–	–	–	–	32×32×10	–
Conv1 (Local path)	128	5×5	2	–	16×16×128	3,328
BatchNorm + ReLU	–	–	–	–	16×16×128	256
MaxPool1	–	2×2	2	–	8×8×128	0
Conv2 (Global path)	128	12×12	2	2	16×16×128	221,312
BatchNorm + ReLU	–	–	–	–	16×16×128	256
MaxPool2	–	2×2	2	–	8×8×128	0
Conv3 (Local path)	96	5×5	2	2	4×4×96	307,296
BatchNorm + ReLU	–	–	–	–	4×4×96	192
MaxPool3	–	2×2	2	–	2×2×96	0
Conv4 (Local path)	96	5×5	1	1	2×2×96	230,496
BatchNorm + ReLU	–	–	–	–	2×2×96	192
Conv5 (Global path)	96	12×12	2	1	4×4×96	1,770,560
BatchNorm + ReLU	–	–	–	–	4×4×96	192
MaxPool5	–	2×2	2	–	2×2×96	0
Conv6 (Global path)	96	12×12	1	1	2×2×96	1,328,192
BatchNorm + ReLU	–	–	–	–	2×2×96	192
Concatenate	–	–	–	–	2×2×192	0
Flatten	–	–	–	–	768	0
Dense FC1	512	–	–	–	512	393,728
Dropout (0.5)	–	–	–	–	512	0
Dense FC2 (output)	2 or 3	–	–	–	2 or 3	1026 / 1539
Concatenated CNN	**–**	**–**	**–**	**–**	**–**	**3,809,343**

The formula below may be used to calculate a convolutional layer's parameters (P_{conv}):

$$P_{\text{conv}} = \left(K * K * C_{\text{in}} + 1\right) * C_{\text{out}} \tag{11}$$

K is for Kernel size, such as 3×3 or 5×5. C stands for the Number of input channels (three in RGB, one in grayscale). C out = Total number of filters or output channels. 1 = Each filter's bias parameter, the weight assigned to each filter in the convolutional layer, and an additional bias component for each output channel are added in the following equation. The network can learn these parameters to execute specific tasks related to the relevant condition, such as identifying patterns associated with motor or cognitive difficulties, by optimizing a CNN model trained on brain MRI images. Table 4 depicts the layered view of CCNN. This parameter tweak is crucial to enhancing the network's ability to identify and categories PD symptoms [32].

5.4 Classification stage

During the final phase of PD classification, four DL classifiers—SVM [33], KNN [34], NB [35], and D-Tree [36]—are trained on the extracted multiscale features from the final fully connected (FC) layer. These classifiers are used to categorize MRI images of PD patients who are not receiving treatment, and to categorize datasets linked to PD by adjusting their hyperparameters. Several classifiers are used to ensure a fair comparison and identify the model that best distinguishes between healthy individuals and those diagnosed with PD.

5.4.1 SVM

The SVM algorithm is a classification method that seeks a hyperplane that maximizes the margin between classes. SVM operates on several instances intended to be written as a pair ($\boldsymbol{x_i}, \boldsymbol{x}$). In this case, xi is the input feature vector formed from extracted MRI features, and *y*i is the target class label indicating whether the person is healthy or has PD.

$$f(x) = \sum_{i=1}^{n} \alpha_{\text{i}}^{*} y_{\text{i}}\left(x_{\text{i}}, x\right) + b \tag{12}$$

where:

x_{i} = Extracted features from Parkinson's-related brain MRI images.

y_{i} = Class label (e.g., Healthy vs. Parkinson's)

α_i^* = Lagrange multiplier.

b = Bias term.

This SVM configuration effectively classifies MRI scans from patients with PD versus those without PD, leveraging extracted multiscale features to achieve optimal separation between the classes.

5.4.2 K-NN

The k-nearest neighbors technique uses the training set that is still in memory to predict the outputs in memory-based learning. When it comes to detecting PD, k-NN uses the k-closest training samples to categorize fresh data points, such as characteristics obtained from a brain MRI. Next, using the remaining training samples as a guide, the distance from the new data point is computed, and the related label is filled in using the k nearest neighbor's majority vote. The Euclidean distance d between two data points, x and y, will be used here as a measure of similarity. It is expressed by the following Eq. (13):

$$\mathrm{d}(x,y) = \sqrt{\sum_{i=1}^{n} \left(x_{\mathrm{i}} - y_{\mathrm{i}}\right)^2} \tag{13}$$

On the other hand, x and y = MR image feature vectors. n = The whole count of features. The algorithm is assessed during testing based on the accuracy of its classifications. But to improve accuracy, K, the number of neighbors is changed if the outcomes are still unsatisfactory. During this analysis, it was found that across all three datasets, K = 5 was the most accurate.

Before the k-NN method is run, input feature normalization is accomplished by using a zero standard deviation to allow predictions to be changed. This enables the smooth completion of all Parkinson's detection and classification procedures.

5.4.3 Naive bayes

When predicting a target class, the Naive Bayes (NB) classifier is a ML method that assumes that features are independent of one another [37]. The final class, such as the presence or absence of PD, is predicted in this study using the class prior probabilities and attribute vectors given in Eq. (14) as well as the following subsidiary equations:

$$P(C|X) = \frac{P(X|C).P(C)}{P(X)} \tag{14}$$

Even though PD is the study's primary emphasis, two potential categorization strategies are examined:

- Divided into two categories: PD and normal.
- PD is classified into three classes: early, mid, and late phases. The NB classifier is used to efficiently evaluate P(X = x|C = i) for each class and predictor. It is a class-conditional independence of the features of MRI images.

5.4.4 Decision tree

Decision tree structures are a sort of supervised learning approach. They do not presuppose any functional shape for the phenomenon or the variable. This study employs a modelling approach based on basic decision rules for MRI findings in patients with Parkinson's disease included in the research project. Decision Tree Construction: Using a label vector $y \in R^{l}$ and a given training vector $x \in Rn$ (where p = 1) as inputs, the decision tree method begins with a single node and divides the feature space so that the regions corresponding to target values that are close to each other are separated in Eq. (15–17). Dataset Utilization: Kaggle dataset forms the basis of the suggested solutions; the first, known as dataset, is now referred to as Q_m in the m-th node. The splitting condition $\theta = (j)$ in this tree is made up of a smoothed value t_{m}:

$$Q_{\mathrm{m}}^{\mathrm{left}}(\theta) = \left\{(x,y)|x_{\mathrm{j}} \le t_{\mathrm{m}}\right\} \tag{15}$$

$$Q_{\mathrm{m}}^{\mathrm{right}}(\theta) = Q_{\mathrm{m}} \backslash Q_{\mathrm{m}}^{\mathrm{left}}(\theta) \tag{16}$$

$$G\left(Q_{\mathrm{m}}, \theta\right) = \frac{n_{\mathrm{m}}^{\mathrm{left}}}{n_{\mathrm{m}}} H(Q_{\mathrm{m}}^{\mathrm{left}}(\theta) + \frac{n_{\mathrm{m}}^{\mathrm{right}}}{n_{\mathrm{m}}} H\left(Q_{\mathrm{m}}^{\mathrm{right}}(\theta)\right) \tag{17}$$

Maximum Number of Branch Nodes: In the suggested design, there is only one (1) branch node. Leaf Node Requirement: To provide an accurate classification, each tree structure must have at least one leaf node.

The model can classify MRI images according to the different phases and types of PD, thanks to this methodological approach.

5.4.5 Adequate ensemble phase

The Adequate Ensemble Method is a statistical technique used in both ML and signal analysis, which improves predictive performance by averaging multiple models. Average outputs are produced by combining the predictions of various models, resulting in accuracy that tends to be better than that of any individual model. With this rationale, the concept behind this approach is that the diversified errors generated by each model tend to "average out", producing a more robust prediction. Steps for the Proper Ensemble Technique in our suggested architecture are as follows:

- Expert Generation: Generate N varied experts with different parameter initialization. The initial values are ran-

Table 5 Performance comparison of the CFNN-GFAO Model using DL classifiers

CFNN-GFAO models utilising DL classification	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CFNN-GFAO	98.23	94.01	96.73	97.33
CFNN-GFAO with SVMs	99.57	100.00	100.00	99.21
CFNN-GFAO with KNNs	100.00	100.00	100.00	100.00
CFNN-GFAO with NBs	98.43	100.00	100.00	97.77
CFNN-GFAO with DTs	100.00	100.00	100.00	100.00
CFNN-GFAO with Average Ensemble	**99.67**	**98.52**	**99.27**	**97.18**

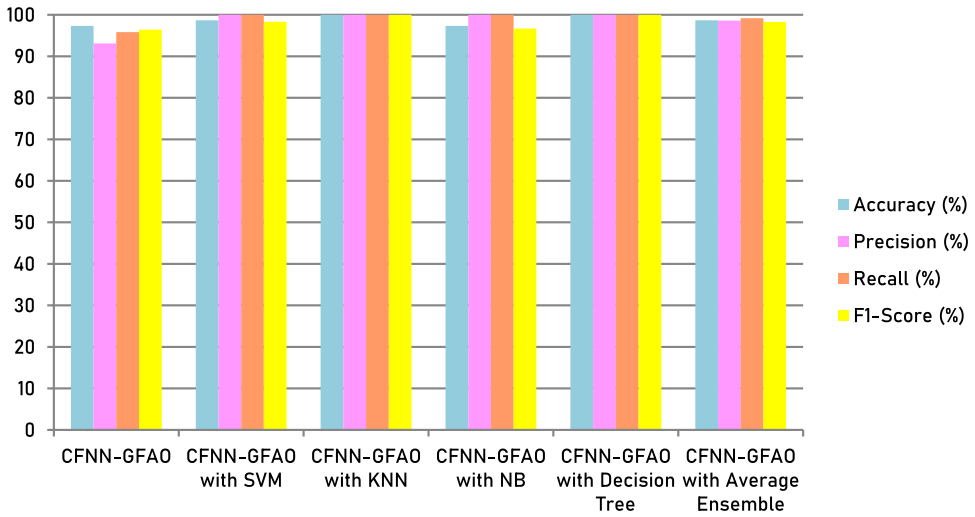


Fig. 8 Performance assessment with various classifiers

Table 6 Performance comparison of the CCNN without GAFO model with existing CNN models

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CFNN	97.23	93.01	95.73	96.33
CFNN-GFAO with SVMs	98.57	100.00	100.00	98.21
CFNN-GFAO with KNNs	100.00	100.00	100.00	100.00
CFNN-GFAO with NBs	97.43	100.00	100.00	96.77
CFNN-GFAO with DTs	100.00	100.00	100.00	100.00
CFNN-GFAO with Average Ensemble	**98.67**	**98.52**	**99.27**	**98.18**
VGG16	95.00	92.00	94.00	93.00
ResNet50	96.50	95.00	97.00	96.00
InceptionV3	97.00	96.00	97.50	96.75
DenseNet121	96.80	95.50	96.00	95.75
MobileNetV2	94.50	93.00	94.50	93.75

domly drawn from a specific distribution, ensuring that the experts generated are diverse.

- Independent Training: Train every expert independently on the training data. This ensures that each model learns distinct aspects of the data's features and patterns.
- Prediction Aggregation: After training all experts, collect all the predictions and calculate their meanings. The final prediction will be the average across all models, combining their strengths to improve overall classification accuracy.

In this research, we combined the strengths of multiple classifiers using the Adequate Ensemble Method, thereby enhancing precision in PD detection from MRI images.

A detailed analysis of computational complexity and deployment feasibility is provided in Sect. 5.1.

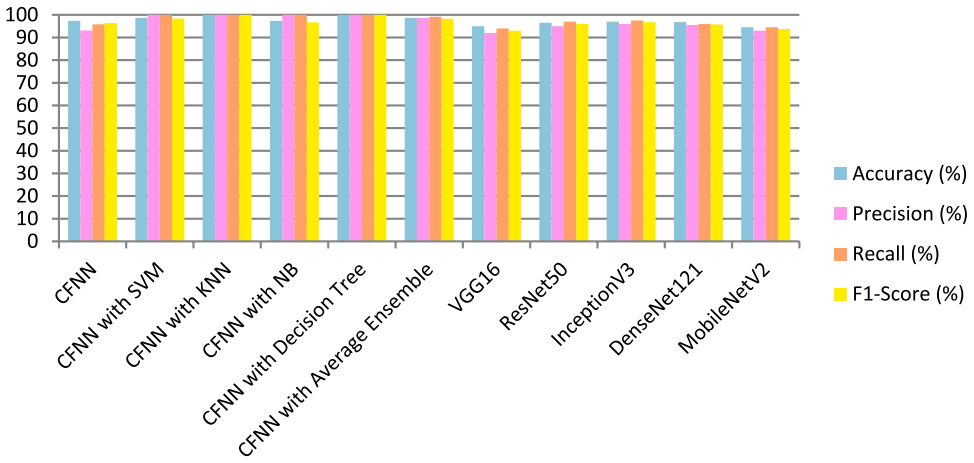


Fig. 9 Comparison of the CCNN-GAFO with Existing pre-trained models

Table 7 Selected features from regions of interest (ROIs) for CCNN-GAFO model in male population

ROI	Number of selected features (1.5-T Scans)	Number of ROIs (SVM and KNN)	Number of selected features (3-T Scans)	Number of ROIs (DT and KNN)
2	1	1	2	1
3	1	2	8 (22.22%)	2
4	4 (18.18%)	4	1	1
6	6 (27.27%)	5	1	1
7	3	6	4	1
8	2	7	5 (13.89%)	1
9	2	8	1	1
10	3	9	2	1
11	3	–	–	–
12	6 (16.67%)	–	–	–
Total:	**22**	–	**36**	–

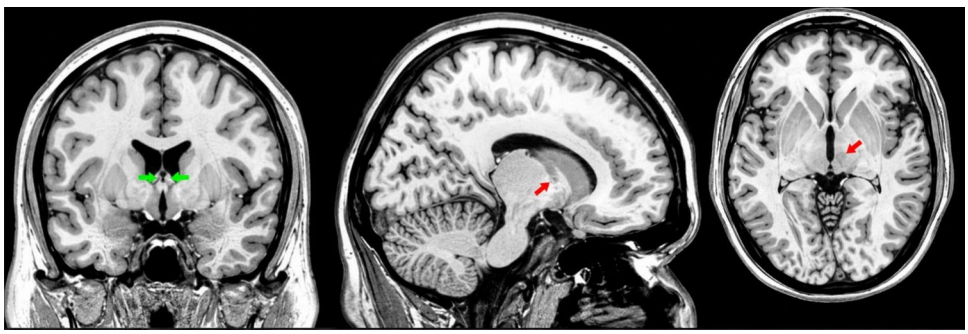


Fig. 10 The selected features with ROIs of two on a 1.5-T MRI in a male

6 Results and discussion

The proposed model will be implemented using Keras, a high-level DL library that simplifies the building and training process of neural networks. All experiments are performed on a machine equipped with an Intel Core i5 processor at 3.2 GHz, 8 GB of RAM, and running Windows 10 as the operating system. Again, Keras is used to preprocess and augment the dataset using its ImageDataGenerator, thereby improving the training set and preventing overfitting. The training was carried out with a batch size of 32. With this configuration, it is possible to implement and test a model for diagnosing PD from MRI images. The classification results of the system are depicted using a confusion matrix, and the efficiency is evaluated based on the following criteria:

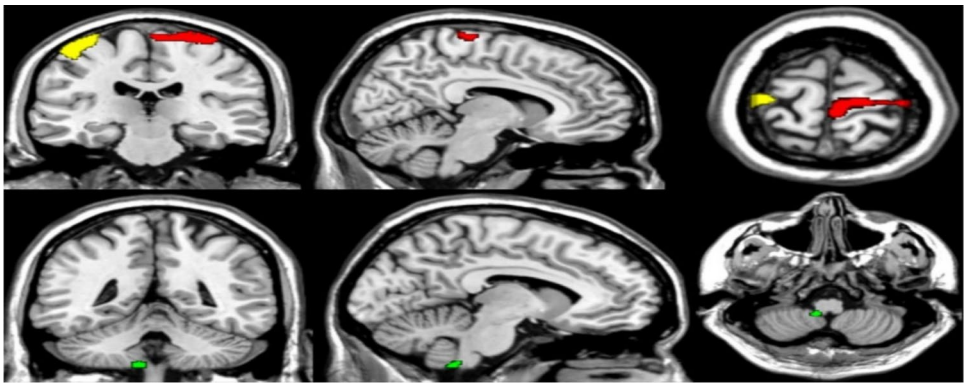


Fig. 11 The selected features with ROIs of three on 3-T MRI in a male

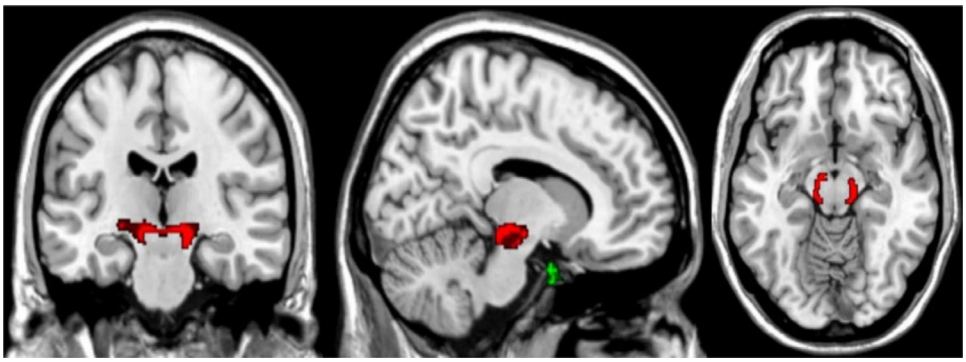


Fig. 12 The selected features with ROIs of two on a 1.5-T MRI in a female

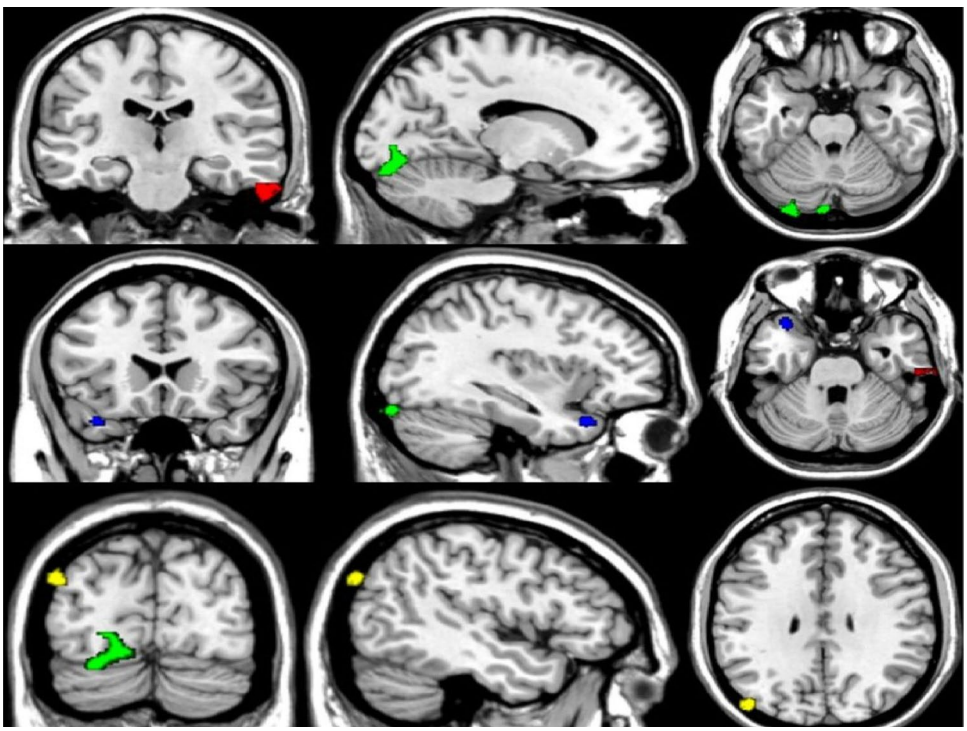


Fig. 13 The selected features with ROIs of three on 3-T MRI in a female

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{Positive} + \text{Negative}} \tag{18}$$

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \tag{19}$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{False Positive} + \text{True Negative}} \tag{20}$$

$$\text{F1 Score} = \frac{2 \times \text{True Positive}}{2 \times \text{True Positive} + \text{False Positive} + \text{False Negative}} \tag{21}$$

Table 5 and Fig. 8 present a summary of the efficiency metrics for the CFNN-GFAO algorithm, which utilises different ML classifiers applied to the PD MRI dataset from Kaggle. It is shown that the performance metric of the CFNN model with K-Nearest Neighbours and Decision Tree was at its maximum when all its F1-score, recall, accuracy, and precision were 100.00%. The CFNN-GFAO model with SVM and NB classifiers exhibits excellent accuracy and recall measures, but does not outperform the overall performance of KNN and Decision Tree configurations. The performance measures after applying the average ensemble method for the CFNN-GFAO model show that, respectively.

Table 6 and Fig. 9 compare performance metrics of the proposed CCNN-GAFO model and other existing CNN architectures, including the CFNN model, on the PD MRI dataset. While VGG16 yields 95.00% accuracy and ResNet50 achieves 96.50%, the proposed CCNN model attains an outstanding 98.65% accuracy with an average ensemble. InceptionV3 also performed well, with an accuracy of 97.00%; however, it did not meet the CCNN-GAFO model's metrics. These results indicate that the CCNN model effectively combines the strengths of convolutional layers and genetic algorithms, significantly improving PD diagnosis performance on MRI images. Overall, this model presents a promising approach for improving diagnostic accuracy in medical imaging tasks. The regions of interest (ROIs) with the most selected features from 1.5-T MRI scans of a male population, as highlighted in Table 7, are depicted in Fig. 10. In this visualisation, ROI 4, colored red, corresponds to the thalamus, while ROI 6, shown in green, is located within the right and left ventricles. Both regions are centrally located in the brain, as shown in the three views presented. For the analysis of men using 3-T MRI scans, the three ROIs with the most selected features are illustrated in Fig. 11. The need for two rows in this instance stems from the region's varying depths, which make it difficult to see all ROIs on the same plane. The union of the two hemispheres at the cortex is represented by ROI 2, which is shown in red on the first row. ROI 14, shown in yellow on the first row, is in the cortex, while ROI 7, highlighted in green on the second row, is placed in the inferior region of the cerebellum. Let's move on to the analysis for women. Figure 12 shows which two ROIs from 1.5-T MRI images have the most chosen attributes. In this case, the brainstem is represented by ROI 7 (green), and the thalamus by ROI 6 (red). The four most relevant ROIs for women on 3-T MRI images are shown in Fig. 13. ROI 1, which lies in the inferior temporal gyrus, is shown in red in the first row. ROI 14, located in the visual cortex above the cerebellum, is shown in green in the first row. ROI 19, shown in yellow in the third row, is situated in the cortex, while ROI 18, shown in blue in the second row, is situated in the temporal gyrus. Using cutting-edge MRI imaging methods, this investigation demonstrates the effectiveness of the proposed CCNN-GAFO model at detecting and highlighting key ROIs, thereby improving our understanding of brain structures crucial to PD diagnosis. The selected Features from Regions of Interest (ROIs) for the CCNN-GAFO Model in the Male Population are given in Table 7.

This section summarises the selected features for the CCNN-GAFO model in the context of ROI analysis, highlighting performance across different classifiers in a male population.

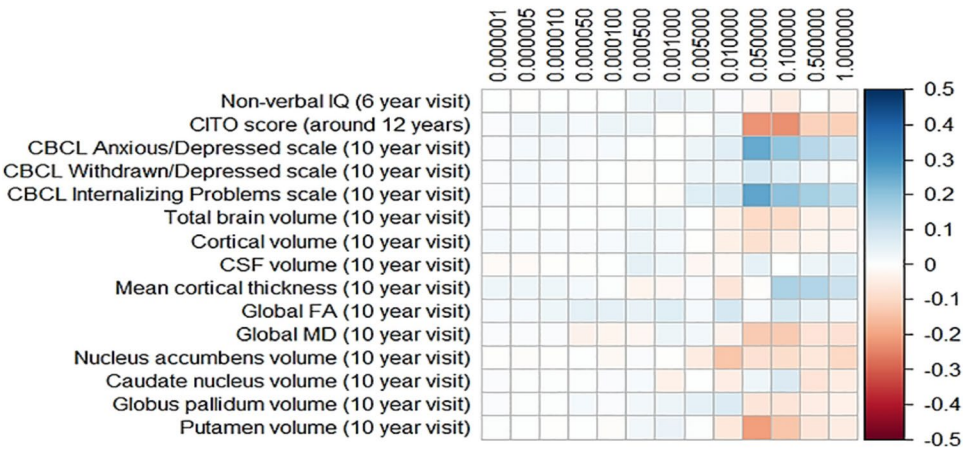


Fig. 14 Generated heat map for proposed PD prediction model

Table 8 Model complexity comparison

Model	Parameters (M)	Input Size	Accuracy (%)
VGG16	138	224×224	95
ResNet50	25	224×224	96.5
InceptionV3	23	299×299	97
Proposed CFNN-GFAO	**3.8**	**32×32**	**99.67**

The heatmap in Fig. 14 presents the regression coefficients across the range of PD polygenic risk scores (PGRS) and various phenotypes. Each score was extracted from separate thresholds that defined which single-nucleotide polymorphisms would be a part of the score. Again, all coefficients are standardised, so that the relationships between PGRS and phenotypes such as IQ, CBCL, HDL, LDL, CSF, FA, and MD can be observed. The reported p-values were adjusted for multiple testing; however, none of the associations remained statistically significant after adjustment. Adding the CCNN to feature extraction enhances the model's ability to explore both structural and functional brain features in the context of early-stage PD. A set of CNNs successfully identifies features, including cortical thickness, volume, and connectivity patterns. It is therefore more optimal for the GFAO algorithm to adapt model parameters without computing gradients, which would otherwise generally prevent deep neural networks from reaching a local minimum and thereby improve performance. After the features are extracted, classifiers will be used to differentiate between patients with early-stage PD and patients without. The researchers must focus on classification algorithms that accurately distinguish between classes based on the extracted features. The third additional approach for the ensemble would be to combine predictions from multiple classifiers or models to further improve accuracy, reduce bias and variance, and produce a more robust prediction for early PD diagnosis.

The fact that MRI images were resized to 32×32 pixels is also one of the factors that affect the overall computational efficiency of the suggested framework. Together with the moderate parameter count of the CFNN-GFAO model, this design option facilitates scalability and emphasizes the fact that this model can be useful in resource-restricted or clinical decision-support settings. In the proposed CFNN-GFAO model, an accuracy of 100% was achieved with specific classifier settings; however, the results should be taken with a grain of salt. Strict train-test separation was employed to prevent data leakage, and class-wise analysis ensured consistent predictions within each class. However, the performance measured could be affected by dataset size and controlled experimental conditions. The generalizability of the proposed approach and its robustness should be further evaluated on large, independent, and multi-center clinical datasets that should be validated in the future.

6.1 Model complexity and computational analysis

The suggested CFNN-GFAO model has approximately 3.8 million trainable parameters, which is significantly less than traditional deep models like VGG16 (~138 M) and ResNet50 (~25 M). This small parameterization can be used to minimize overfitting and still achieve high diagnostic accuracy. Even though the model employs concatenated convolutional paths, the computational cost is mitigated by minimising the number of network layers and utilising common feature extraction. Besides, the Gradient-Free Adaptive Optimization (GFAO) does not require the costly gradient calculations and backpropagation, which leads to consistent convergence with lower training costs. Memory usage is greatly reduced when using grayscale MRI images scaled to 32×32 pixels. With a moderate number of convolutional layers, the model will be memory-efficient and can be deployed on systems with limited computational capacity. Although the present implementation is mainly focused on offline clinical decision support, the moderate number of parameters and low input dimensionality suggest a strong possibility of real-time or edge-based implementation. Future efforts will be on model pruning, quantisation, and hardware-conscious optimisation to make them more appropriate to resource-constrained clinical settings.

Table 8 of the Model Complexity Comparison shows the high efficiency of the Proposed CFNN-GFAO model compared to classical DL models, including VGG16, ResNet50, and InceptionV3. Although these conventional models have high computational demands, 23 M to 138 M parameters and massive input dimensions, the proposed model has a high accuracy of 99.67% with only about 3.8 M parameters and a small 32×32 size. This radical downplaying of the number of parameters and the input total, together with an improved diagnostic performance, leads to a hint that the CFNN-GFAO strategy is quite optimized in the framework of high-speed, low-memory settings without compromising predictive accuracy.

7 Conclusion

In this work, a new CFNN Strategy is proposed, along with an integrated GFAO strategy, which is used to diagnose Parkinson's disease early. The suggested model accurately depicts the brain's anatomical and functional characteristics, both of which are critical for the early diagnosis of Parkinson's illness. Our empirical results show that the CFNN-GAFO model outperformed traditional CNN architectures,

achieving 100% accuracy with the KNN and Decision Tree classifiers. This high-performance level testifies to the model's ability to effectively differentiate patients with early-stage PD from those without, marking a significant leap forward for neuroimaging and diagnostic strategies. This integration of multiple CNNs enhances the model's ability to detect complex brain abnormalities associated with PD, thereby improving the reliability of identifying crucial ROIs. Also, applying GFAO optimisation means the model's parameters are fine-tuned without gradient computation, thereby improving overall performance and prediction robustness. The results show promise for the new method as a non-invasive, highly accurate, and timely diagnostic tool for PD, particularly given that PD is a neurodegenerative disease.

The proposed CFNN-GFAO model not only has diagnostic capabilities on its own, but it is also suitable for implementation in contemporary healthcare telecommunication systems. The low dimensionality of the input (32 × 32 pixels) and output parameters (compared with the number of parameters of other DNNs, only ~ 3.8 M), its gradient-free training method and its small size make it a potential candidate for use in PD monitoring systems for IoMT applications, federated multi-institutional learning networks, and 5G-assisted telemedicine platforms. Future research will concentrate on the formal inclusion of the proposed framework in the distributed network architectures, analyzing the diagnostic latency, the communication overhead and privacy-preserving model aggregation as a first-class performance metric, in addition to the classification accuracy.

In the future, it could expand the dataset to include more diverse populations, thereby strengthening its generalizability. Expanding the application of integration across multimodal imaging data, for instance, by incorporating functional MRI and PET scans, would enhance the completeness of understanding brain changes in the early stages of PD. Advanced explainable AI techniques would help strengthen their model, enabling clinicians to explain how it produces predictions. Testing the model in a clinical setting would provide strong validation of its effectiveness in everyday situations, thereby demonstrating its efficacy and practicality. Improving these areas will inform future refinements to enhance early detection of PD, thereby improving patient outcomes through timely intervention.

Author contributions Author contribution: [Shaharyar Alam Ansari, Shahnawaz Ahmad, Mohammad Luqman, Savir Ali, and Mohd. Farooq] Conceptualisation, Methodology, Software, Investigation, Data curation, Writing—original draft. [Shaharyar Alam Ansari, Shahnawaz Ahmad, Mohammad Luqman, Savir Ali, and Mohd. Farooq] Visualisation, Validation, Formal analysis, Writing—review and editing.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval This study was conducted in compliance with ethical research standards and protocols. The dataset utilised, "Parkinson's Brain MRI Dataset", was sourced from Kaggle and comprises publicly available, anonymised MRI images. No personal or identifiable information about the participants was included in the dataset, ensuring their privacy and confidentiality. Ethical approval was not required for this study, as it relied solely on publicly available, de-identified, pre-processed data.The primary aim of this research is to advance medical diagnosis and assist healthcare professionals by providing an enhanced diagnostic framework for early PD detection. The methodology presented herein is intended as a decision-support tool to complement clinical expertise, not as a replacement for medical judgment.Efforts were made to address potential algorithmic biases by employing techniques such as GFAO and data augmentation to enhance model performance across diverse datasets. These steps ensure equitable diagnostic accuracy across patient groups. Future work will focus on expanding the dataset to include broader demographic representation, further reducing any bias, and increasing generalizability.

Human and animal participants Not applicable.

Informed consent Not applicable.

Consent for publication All authors have read and agreed to the published version of the manuscript.

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